

UGB Toppers' Mock Test-3

Solutions

Problem 1

Prove that there exists an infinite sequence of positive integers $\{a_n\}_{n \geq 1}$ such that $a_1^2 + a_2^2 + \dots + a_n^2$ is a perfect square for every $n \geq 1$.

Solution.

We construct the sequence recursively. Write $S_n^2 = \sum_{k=1}^n a_k^2$, and require S_n to be an odd positive integer at every stage.

Suppose $a_1, \dots, a_n$ have been chosen so that S_n is odd. We seek a positive integer a_{n+1} for which $S_n^2 + a_{n+1}^2$ is again a perfect square. Setting $S_{n+1}^2 = S_n^2 + a_{n+1}^2$ gives

$$a_{n+1}^2 = (S_{n+1} - S_n)(S_{n+1} + S_n).$$

Imposing $S_{n+1} - S_n = 1$ forces $a_{n+1}^2 = 2S_n + 1$. Equivalently, take

$$a_{n+1} = \frac{S_n^2 - 1}{2}, \quad S_{n+1} = \frac{S_n^2 + 1}{2}.$$

Base case. Set $a_1 = 3$, so $S_1 = 3$, which is odd.

Inductive step. If $S_n = 2k + 1$ with $k \geq 1$, then

$$a_{n+1} = \frac{(2k+1)^2 - 1}{2} = 2k^2 + 2k,$$

a positive integer. The new partial sum satisfies

$$S_n^2 + a_{n+1}^2 = S_n^2 + \left(\frac{S_n^2 - 1}{2}\right)^2 = \frac{S_n^4 + 2S_n^2 + 1}{4} = \left(\frac{S_n^2 + 1}{2}\right)^2,$$

so $S_{n+1} = \frac{1}{2}(S_n^2 + 1) = 2k^2 + 2k + 1$, which is again odd. The recursion therefore continues indefinitely, producing the required sequence. $\square$

Problem 2

Let $z_1, z_2, \dots, z_n$ be non-zero complex numbers, all lying strictly on one side of some straight line through the origin of the complex plane. Prove that $\sum_{k=1}^n z_k \neq 0$ and $\sum_{k=1}^n \frac{1}{z_k} \neq 0$.

Solution.

Because all the z_k lie strictly on one side of a line through the origin, a suitable rotation carries that line onto the imaginary axis and sends every z_k into the open right half-plane. Concretely, there is a unit complex number c such that $z_k = c w_k$ with $\operatorname{Re}(w_k) > 0$ for every k .

First sum. We have

$$\sum_{k=1}^n z_k = c \sum_{k=1}^n w_k,$$

and

$$\operatorname{Re}\left(\sum_{k=1}^n w_k\right) = \sum_{k=1}^n \operatorname{Re}(w_k) > 0.$$

Hence $\sum w_k \neq 0$, and multiplication by the non-zero constant c gives $\sum z_k \neq 0$.

Second sum. Compute

$$\sum_{k=1}^n \frac{1}{z_k} = \frac{1}{c} \sum_{k=1}^n \frac{1}{w_k} = \frac{1}{c} \sum_{k=1}^n \frac{\overline{w_k}}{|w_k|^2}.$$

The real part of the sum on the right is

$$\sum_{k=1}^n \frac{\operatorname{Re}(w_k)}{|w_k|^2} > 0,$$

since each summand is strictly positive. The sum is therefore non-zero, and multiplication by $1/c$ shows $\sum 1/z_k \neq 0$. $\square$

Problem 3

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Suppose $f(a) = f(b) = 0$ and f is not identically zero on $[a, b]$. Prove that for every positive integer n there exists $c \in (a, b)$ such that $(b - c)f'(c) = n f(c)$.

Solution.

Define

$$g(x) = f(x)(b - x)^n.$$

Then g is continuous on $[a, b]$ and differentiable on (a, b) , with

$$g(a) = f(a)(b - a)^n = 0, \quad g(b) = f(b) \cdot 0 = 0.$$

By Rolle's theorem there exists $c \in (a, b)$ with $g'(c) = 0$. Differentiating gives

$$g'(x) = f'(x)(b - x)^n - n f(x)(b - x)^{n-1} = (b - x)^{n-1}[(b - x)f'(x) - n f(x)].$$

Since $c < b$, the factor $(b - c)^{n-1}$ is non-zero, so

$$(b - c)f'(c) - n f(c) = 0,$$

which is the required identity. $\square$

Problem 4

Let $f : [0, 2\pi] \rightarrow \mathbb{R}$ be a continuous function, not identically zero, with

$$\int_0^{2\pi} f(x) dx = \int_0^{2\pi} f(x) \cos x dx = \int_0^{2\pi} f(x) \sin x dx = 0.$$

Prove that f has at least three distinct zeros in $(0, 2\pi)$.

Solution.

We argue by contradiction. Since $\int_0^{2\pi} f = 0$ and f is continuous and not identically zero, f must change sign in $(0, 2\pi)$; otherwise the integral would be strictly positive or strictly negative. So f has at least one sign change.

Suppose, for contradiction, that f changes sign at exactly two points $x_1 < x_2$ in $(0, 2\pi)$. Consider the trigonometric polynomial of degree one

$$T(x) = \cos\left(x - \frac{x_1+x_2}{2}\right) - \cos\left(\frac{x_2-x_1}{2}\right).$$

The equation $T(x) = 0$ reduces to $x - \frac{x_1+x_2}{2} \equiv \pm \frac{x_2-x_1}{2} \pmod{2\pi}$, whose only solutions in $(0, 2\pi)$ are x_1 and x_2 . Moreover T changes sign at each of these points and nowhere else.

Since f and T change sign at exactly the same two points, the product $f(x)T(x)$ has constant sign throughout $(0, 2\pi)$. As f is continuous and not identically zero, fT is not identically zero, and therefore

$$\int_0^{2\pi} f(x)T(x) dx \neq 0.$$

On the other hand, expanding the cosine yields

$$T(x) = c_1 + c_2 \cos x + c_3 \sin x$$

for some real constants c_1, c_2, c_3 . The hypothesis on f then gives

$$\int_0^{2\pi} f(x)T(x) dx = c_1 \int_0^{2\pi} f + c_2 \int_0^{2\pi} f \cos x + c_3 \int_0^{2\pi} f \sin x = 0,$$

a contradiction.

A sign change at exactly one point is excluded by the same argument applied to $T(x) \equiv 1$, since then $\int_0^{2\pi} f$ could not vanish. Therefore f has at least three sign changes, hence at least three distinct zeros in $(0, 2\pi)$. $\square$

Problem 5

Let $ABCD$ be a convex quadrilateral inscribed in a parabola. The tangents to the parabola at A, B, C, D meet pairwise at P, Q, R, S , where $P = \tan_A \cap \tan_B$, $Q = \tan_B \cap \tan_C$, $R = \tan_C \cap \tan_D$, $S = \tan_D \cap \tan_A$. Prove that $[ABCD] = 2[PQRS]$.

Solution.

Place the parabola in standard position as $y = x^2$, and let

$$A(a, a^2), \quad B(b, b^2), \quad C(c, c^2), \quad D(d, d^2)$$

with $a < b < c < d$, so that $ABCD$ is convex.

By the shoelace formula,

$$2[ABCD] = |ab(b-a) + bc(c-b) + cd(d-c) + da(a-d)|.$$

The tangent at (t, t^2) has equation $y = 2tx - t^2$. Intersecting the tangents at A and B gives

$$2ax - a^2 = 2bx - b^2 \implies x = \frac{a+b}{2}, \quad y = ab.$$

By symmetry the four vertices of $PQRS$ are

$$P\left(\frac{a+b}{2}, ab\right), \quad Q\left(\frac{b+c}{2}, bc\right), \quad R\left(\frac{c+d}{2}, cd\right), \quad S\left(\frac{d+a}{2}, da\right).$$

Applying the shoelace formula again,

$$2[PQRS] = |(x_P y_Q - y_P x_Q) + (x_Q y_R - y_Q x_R) + (x_R y_S - y_R x_S) + (x_S y_P - y_S x_P)|.$$

The first cross-term simplifies as

$$x_P y_Q - y_P x_Q = \frac{a+b}{2} \cdot bc - ab \cdot \frac{b+c}{2} = \frac{b^2(c-a)}{2},$$

and the remaining three terms are obtained by cyclic shift:

$$4[PQRS] = |b^2(c-a) + c^2(d-b) + d^2(a-c) + a^2(b-d)|.$$

Expanding and regrouping,

$$b^2(c-a) + c^2(d-b) + d^2(a-c) + a^2(b-d) = -[ab(b-a) + bc(c-b) + cd(d-c) + da(a-d)].$$

Thus $4[PQRS] = 2[ABCD]$, which gives $[ABCD] = 2[PQRS]$. $\square$

Problem 6

Let ABC be a triangle with angles A, B, C . Prove that

$$\frac{\cos A}{\sin^3 A} + \frac{\cos B}{\sin^3 B} + \frac{\cos C}{\sin^3 C} \geq \frac{3}{2 \sin A \sin B \sin C}.$$

Solution.

Set $x = \cot A$, $y = \cot B$, $z = \cot C$. The triangle identity $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$ gives the constraint $xy + yz + zx = 1$.

Each term on the left becomes

$$\frac{\cos A}{\sin^3 A} = \cot A \cdot \csc^2 A = \cot A (1 + \cot^2 A) = x + x^3,$$

so writing $S = x + y + z$ and $P = xyz$,

$$\text{LHS} = (x + y + z) + (x^3 + y^3 + z^3) = S + [S^3 - 3S(xy + yz + zx) + 3P] = S^3 - 2S + 3P.$$

For the right-hand side, the identity $\cot A + \cot B + \cot C - \cot A \cot B \cot C = \csc A \csc B \csc C$ gives

$$\text{RHS} = \frac{3}{2}(S - P).$$

The inequality is therefore equivalent to

$$S^3 - 2S + 3P \geq \frac{3}{2}(S - P) \iff 2S^3 - 7S + 9P \geq 0.$$

Now introduce $u = x + y$, $v = y + z$, $w = z + x$. In an acute, right, or obtuse triangle (the only cases) one checks that

$$x + y = \cot A + \cot B = \frac{\sin C}{\sin A \sin B} > 0,$$

so $u, v, w > 0$. Moreover $S = \frac{1}{2}(u + v + w)$ and a direct expansion using $xy + yz + zx = 1$ gives

$$S - P = (x + y)(y + z)(z + x) = uvw, \quad S^2 + 1 = uv + vw + wu.$$

Substituting $9P = 9S - 9uvw$ into the inequality,

$$2S^3 - 7S + 9P = 2S^3 + 2S - 9uvw = 2S(S^2 + 1) - 9uvw = (u + v + w)(uv + vw + wu) - 9uvw.$$

By AM-GM, $u + v + w \geq 3\sqrt[3]{uvw}$ and $uv + vw + wu \geq 3\sqrt[3]{u^2v^2w^2}$; multiplying gives

$$(u + v + w)(uv + vw + wu) \geq 9uvw,$$

so $2S^3 - 7S + 9P \geq 0$, which is the required inequality. Equality holds when $u = v = w$, i.e. when the triangle is equilateral. $\square$

Problem 7

Let $f(x)$ be a real polynomial of degree at most $2n$ such that $|f(k)| \leq 1$ for every integer k with $-n \leq k \leq n$. Prove that $|f(x)| \leq 2^{2n}$ for every real x with $-n \leq x \leq n$.

Solution.

Since f has degree at most $2n$, it is determined by its values at the $2n + 1$ integer nodes $-n, -n + 1, \dots, n$. The Lagrange interpolation formula gives

$$f(x) = \sum_{k=-n}^n f(k) \prod_{\substack{j=-n \\ j \neq k}}^n \frac{x - j}{k - j}.$$

For $-n \leq k \leq n$,

$$\prod_{\substack{j=-n \\ j \neq k}}^n (k-j) = (-1)^{n-k} (n+k)! (n-k)!,$$

so taking absolute values and using $|f(k)| \leq 1$,

$$|f(x)| \leq \sum_{k=-n}^n \frac{\prod_{j \neq k} |x-j|}{(n+k)! (n-k)!} = \sum_{k=-n}^n \binom{2n}{n+k} \cdot \frac{\prod_{j \neq k} |x-j|}{(2n)!}.$$

It remains to bound the product $\prod_{j \neq k} |x-j|$ for $x \in [-n, n]$. The values $|x-j|$ as j ranges over the $2n$ integers in $\{-n, \dots, n\} \setminus \{k\}$ are $2n$ non-negative reals. Sort them as $d_1 \leq d_2 \leq \dots \leq d_{2n}$.

We claim $d_i \leq i$ for each i . Indeed, $d_i \leq i$ is equivalent to saying that at least i of the integers $j \in \{-n, \dots, n\} \setminus \{k\}$ lie within distance i of x . The closed interval $[x-i, x+i]$ has length $2i$ and so contains at least $2i$ consecutive integers; removing the single excluded value k leaves at least $2i-1 \geq i$ of them (for $i \geq 1$). This proves the claim.

Hence

$$\prod_{j \neq k} |x-j| = \prod_{i=1}^{2n} d_i \leq \prod_{i=1}^{2n} i = (2n)!,$$

and therefore

$$|f(x)| \leq \sum_{k=-n}^n \binom{2n}{n+k} = \sum_{m=0}^{2n} \binom{2n}{m} = 2^{2n}. \quad \square$$

Problem 8

Let p be an odd prime. Prove that

$$\sum_{j=0}^p \binom{p}{j} \binom{p+j}{j} \equiv 2^p + 1 \pmod{p^2}.$$

Solution.

Split the sum

$$S = \sum_{j=0}^p \binom{p}{j} \binom{p+j}{j}$$

into the contributions from $j = 0$, $1 \leq j \leq p-1$, and $j = p$.

The term $j = 0$. This contributes $\binom{p}{0} \binom{p}{0} = 1$.

The middle terms $1 \leq j \leq p-1$. For such j the identity $\binom{p}{j} = \frac{p}{j} \binom{p-1}{j-1}$ shows that $\binom{p}{j}$ is divisible by p . Next,

$$\binom{p+j}{j} = \prod_{k=1}^j \frac{p+k}{k} = \prod_{k=1}^j \left(1 + \frac{p}{k}\right) \equiv 1 + p \sum_{k=1}^j \frac{1}{k} \pmod{p^2},$$

where the inverses are taken modulo p inside the term of order p . Multiplying,

$$\binom{p}{j} \binom{p+j}{j} \equiv \binom{p}{j} + p \binom{p}{j} \sum_{k=1}^j \frac{1}{k} \pmod{p^2}.$$

The second term contains the factor $p \binom{p}{j}$, which is divisible by p^2 , so

$$\binom{p}{j} \binom{p+j}{j} \equiv \binom{p}{j} \pmod{p^2}.$$

The term $j = p$. Here $\binom{p}{p} \binom{2p}{p} = \binom{2p}{p}$. Vandermonde's identity gives

$$\binom{2p}{p} = \sum_{k=0}^p \binom{p}{k}^2 = 1 + \sum_{k=1}^{p-1} \binom{p}{k}^2 + 1.$$

For $1 \leq k \leq p-1$ the coefficient $\binom{p}{k}$ is divisible by p , so $\binom{p}{k}^2$ is divisible by p^2 . Hence

$$\binom{2p}{p} \equiv 2 \pmod{p^2}.$$

Combining the parts.

$$S \equiv 1 + \sum_{j=1}^{p-1} \binom{p}{j} + 2 \pmod{p^2}.$$

Since $\sum_{j=0}^p \binom{p}{j} = 2^p$, the middle sum equals $2^p - 2$, and therefore

$$S \equiv 1 + (2^p - 2) + 2 \equiv 2^p + 1 \pmod{p^2}. \quad \square$$